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Chapter 8 - Functions & Recursions:

A function is a group of statements performing a specific task.

When a program gets bigger in size and its complexity grows, it gets difficult for a programmer to keep track on which piece of code is doing what!

A function can be reused by the programmer in a given program any number of.

Example and Syntax of a function

The syntax of a function looks as follows:

```
def func 1 ():
    print ('Hello')
```

This function can be called any number of times, anywhere in the program.

Function Call :

Whenever we want to call a function we put the name of the function followed by parenthesis as follows:

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func1() → This is called function call.

function definition :

The part containing the exact set of instructions which are executed during the function call.

Quick Quz : Write a program to greet a user with "good day" using functions.

Sol :
```
def func():
    name = str(input("enter your name : "))
    print ("hood day", name)

Func ()
Func ()
```

Types of functions in python

There are two types of functions in Python:
1> Built in functions → Already Present in Python
2> User defined functions → Defined by the user

- Examples of built in function inclu[redacted] len() , print(), range() etc.

- The ~~function~~ func1() we defined is an example of user defined function.

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functions with arguments :
A function can accept some values it can work with. We can put these values in the parenthesis. A function can also return values as shown below:

```
def greet (name):
        gr = "Hello "+name
        return gr
```

"Harry" is passed to greet in name.

```
a = greet ("Harry")
```

↳ a will now contain "Hello Harry".

Default Parameter value :

We can have a value as default argument in a function.
If we specify name = "strange" in the line containing def, this value is used when no argument is passed.

for example :

```
def greet (name = "strange"):
        # function body
```

greet () → Name will be "strange" in function body (de
greet ("Harry") → Name will be "Harry" in function body (passed)

Recursion :
Recursion is a function which calls itself

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It is used to directly use a mathematical formula as a function for example :

factorial (n) = n x factorial (n-1)

This function can be defined as fallows:

```
def factorial (n):
    if i == 0 or i == 1 : -> Base condition which does
        return                 call the function any further
    else
        return n* factorial (n-1) -> function calling
                                     itself.
```

This works as fallows:

factorial (4) → (function called)
4 x factorial (3)
4 x [3 x factorial (2)]
4 x 3 x [2 factorial (1)]
4 x 3 x 2 x [1] [function returned]

Note:
The programmer need to be extremely careful while working with recursion to ensure that the function does not Infinitely keep calling itself. Recursion is Sometimes the most direct way to code an algo-rithm.